



Chapter:- BIOMOLECULE



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- Q1-** Glucose or sucrose are soluble in water but cyclohexane or benzene (simple six membered ring compounds) are insoluble in water. Explain.
- Q2-** What are the expected products of hydrolysis of lactose?
- Q3-** How do you explain the absence of aldehyde group in the pentaacetate of D-glucose?
- Q4-** The melting points and solubility in water of amino acids are generally higher than that of the corresponding halo acids. Explain.
- Q5-** Where does the water present in the egg go after boiling the egg?
- Q6-** Why cannot vitamin C be stored in our body?
- Q7-** What products would be formed when a nucleotide from DNA containing thymine is hydrolysed?
- Q8-** When RNA is hydrolysed, there is no relationship among the quantities of different bases obtained. What does this fact suggest about the structure of RNA?
- Q9-** What are monosaccharides?
- Q10-** What are reducing sugars?
- Q11-** Write two main functions of carbohydrates in plants.
- Q12-** Classify the following into monosaccharides and disaccharides. Ribose, 2-deoxyribose, maltose, galactose, fructose and lactose.
- Q13-** What do you understand by the term glycosidic linkage?
- Q14-** What is glycogen? How is it different from starch?
- Q15-** What are the hydrolysis products of
(i) sucrose and (ii) lactose?
- Q16-** What is the basic structural difference between starch and cellulose?
- Q17-** What happens when D-glucose is treated with the following reagents?
(i) HI (ii) Bromine water (iii) HNO_3
- Q18-** Enumerate the reactions of D-glucose which cannot be explained by its open chain structure.
- Q19-** What are essential and non-essential amino acids? Give two examples of each type.
- Q20-** Define the following as related to proteins
(i) Peptide linkage (ii) Primary structure (iii) Denaturation.
- Q21-** What are the common types of secondary structure of proteins?
- Q22-** What type of bonding helps in stabilising the α -helix structure of proteins?
- Q23-** Differentiate between globular and fibrous proteins.
- Q24-** How do you explain the amphoteric behaviour of amino acids?



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- Q25- What are enzymes?
- Q26- What is the effect of denaturation on the structure of proteins?
- Q27- How are vitamins classified? Name the vitamin responsible for the coagulation of blood.
- Q28- Why are vitamin A and vitamin C essential to us? Give their important sources.
- Q29- What are nucleic acids? Mention their two important functions.
- Q30- What is the difference between a nucleoside and a nucleotide?
- Q31- The two strands in DNA are not identical but are complementary. Explain.
- Q32- Write the important structural and functional differences between DNA and RNA.
- Q33- What are the different types of RNA found in the cell?